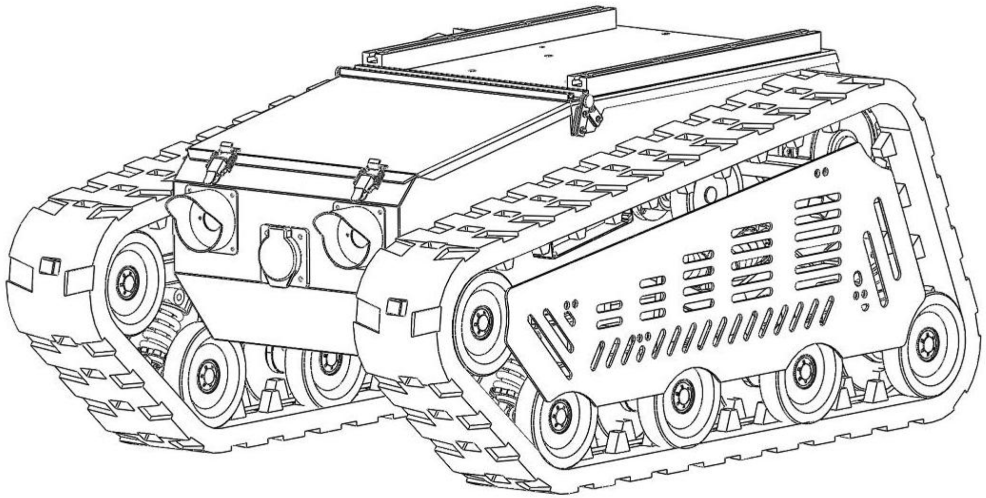
# Operation Manual

(TinS-8 Robot Chassis)



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**Version 1.0**

## 1. Introduction:

1) The overall track structure of the chassis is designed in a parallelogram shape, with high ground clearance, strong crossing ability, long track grounding area, excellent stair climbing ability, and strong outdoor off-road ability. Both the front and rear tracks can wrap the vehicle body, which is more conducive to protecting the vehicle from damage and allowing it to walk on various terrains.

2) Protection level: IP65.

3) Communication protocol: 485 communication, CAN protocol.

4) Working mode: DC brushless Hall motor or servo motor drive+encoder, closed-loop control system, precise control. Precise control of steering angle, speed, and stroke. It can rotate in place or spiral as needed.

5) Driver: overvoltage, overcurrent, undervoltage, overheating, short circuit protection.

6) Control methods: joystick control, standard aircraft model remote control, and upper computer command control. Speed loop+displacement loop+differential loop, precise speed regulation.

7) Rubber tracks: good ground adhesion, long track grounding area.

8) Tensioning wheel: adjustable to ensure normal tensioning and operation of the track.

9) Widely applied in fields such as safety protection monitoring, robotic arm carriers, lifting platforms, transportation vehicle, etc.

10) Lithium iron phosphate battery: It has the advantages of high working voltage, high energy density, long cycle life, good safety performance, low self discharge rate, and no memory effect.

11) Shock absorption system: Matilda suspension combined with Christie shock absorption, providing excellent shock absorption system.

12) Switching control of high and low beams allows for night driving operations.

13) Support customization and secondary development.

14) Optional powerful dual degree of freedom Pan/Tilt rotating platform, horizontal direction: torque 100NM, speed 180 °/s, vertical direction:

torque 100NM, speed 180 °/s or torque 50NM, speed 360 °/s.

## 2. Chassis Specification:

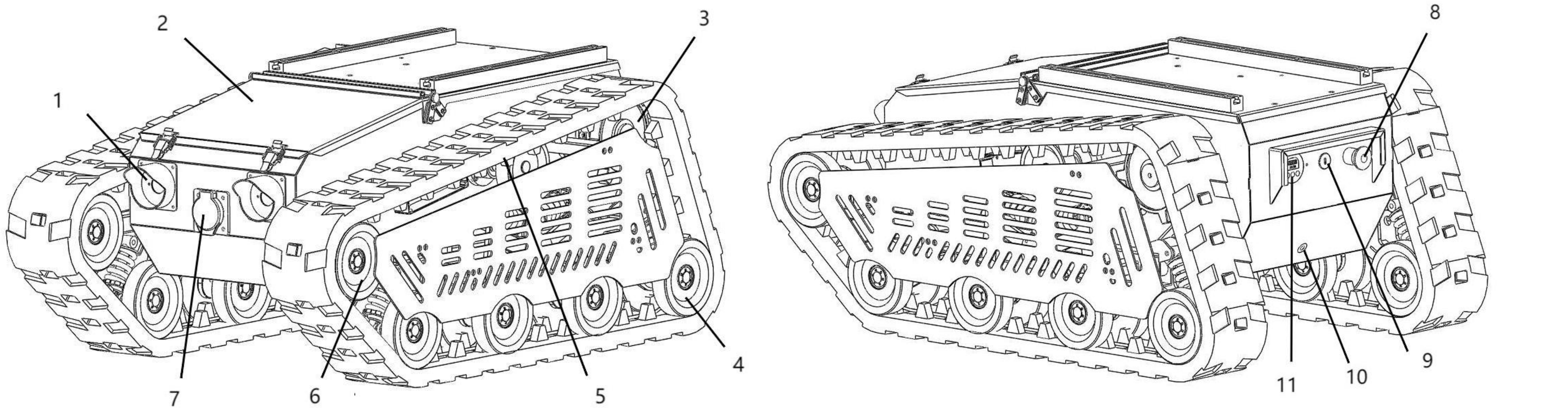
| Item | Data | Item | Data |
|---|---|---|---|
| Dimension | 1060*672*465mm | Inner battery space | 360*330*220MM |
| Self Weight | 160kg (include battery) | Speed | 0-8km/h |
| Waterproof | IP65 | Designed Load | 100kg |
| Track Width | 130mm | Rated Torque | 40NM*2 |
| Gear Ratio | 1:10 | Climbing Ability | ≤30°(can climb stairs) |
| Motor | 1.5KW*2 | Motor Rated Speed | 3000r/min |
| Ground clearance | 150mm | Maximum Span | 230mm |
| Battery | 48v 50AH Lithium iron phosphate battery 280*260*220mm, 25kg | Continuous working time | 3-4h |
| Obstacle crossing ability | 230mm | Track grounding length | 715mm |
| Battery discharging current (only for chassis) | 100A | Voltage | 48v |

**Packing list**

| No. | Item | Qty. | Note |
|---|---|---|---|

| | | | |
|---|---|---|---|
| 1 | Chassis | 1 | |
| 2 | Remote controller | 1 | |
| 3 | Tool | 1 | |
| 4 | Battery | 1 | Optional |
| 5 | Battery charger | 1 | Optional |

### 3 .Model and composition




1. Headlight 2 Upper cover 3 Drive wheel 4 Load-bearing wheel 5 Rubber track 6 Tensioning wheel 7. Charging port

8 Emergency stop switch 9. Power switch 10 Oil drain hole 11 Power display

**Operation steps:**

1. Take out the robot and remote controller.

2. First, turn on the power switch on the chassis and check if the battery is fully charged.

3. Turn on the switch on the remote controller and start connecting the chassis.

4. After successfully connecting the chassis, the remote controller can be used to control the forward and backward movement of the chassis.

## 4.Interface Definition Description

1. Definition of chassis external expansion interface

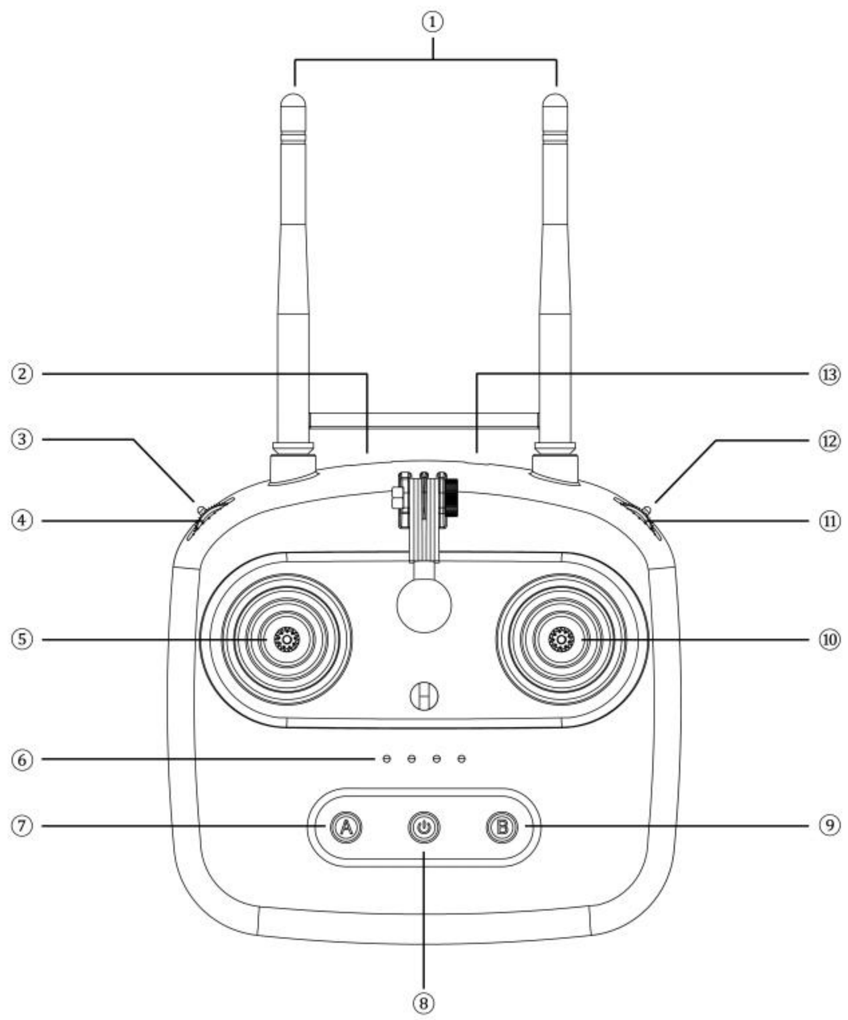
    1. 48V, 24V DC power interface

    2. 485 communication interface

2. Definition of driver interface

Please refer to the chassis wiring diagram definition

## 5. Instructions for the remote controller



1. Right stick 10　X2, Y2 channels control the forward, backward and speed of the chassis

2. DIP switch 3 channel E controls the chassis far and near beams

3. The remote control is currently equipped with 10 channels for use (4 channels for the chassis)

4. Reserve 6 control channels, which are suitable for the control end of the user's secondary development

5. For details, please refer to the T10 remote control manual and operate it carefully, or consult our company for technical assistance.

## 6. Cautions

1. Please charge the battery in time to avoid overshoot and overdischarge.

2. Please refer to the product manual in detail before use to avoid unnecessary accidents.

3. Before operating the equipment for the first time, please try to operate proficiently on a wide ground to avoid accidents caused by unsuitable controls.

4. If the fault cannot be solved independently, please contact us for technical support.

## 7. Common faults and treatment methods

| No. | Fault phenomenon | Reason for malfunction | maintenance methods |
|---|---|---|---|
| 1 | The whole machine is without power | The power socket is out of power | Test pen to check power supply |

| | | | |
|---|---|---|---|
| 2 | | The power cord is not securely plugged in | Tighten the power plug |
| 3 | | The switch power supply is not turned on | Insert the key and turn it clockwise to turn on the power |
| 6 | Robots do not walk or cannot walk straight | Walking motor malfunction | Open the cover of the walking robot, operate the joystick, observe the operation of the two motors, and perform maintenance |
| 7 | No display on the monitor | The power switch of the monitor does not light up | Press the power switch on the monitor |

**8.Instructions and precautions for using lithium iron phosphate batteries**

The battery configured for this product is a lithium iron phosphate battery. When using a lithium iron phosphate battery, the following points should be noted to ensure the performance and lifespan of the battery:

(1) Charging precautions:

1. If the battery level displays below 48V or the battery level display alarms, the battery should be charged in a timely manner. After plugging in the charger, it starts working and the charging indicator light flashes red, indicating that the charging is normal. After the battery is fully charged, the green indicator light will turn on, indicating that the charging work is completed.

2. Use the original standard charger for charging, avoid using inferior or other types of chargers to prevent damage to the battery pack. The charging time should not exceed 24 hours to avoid the battery being in a float charging state for a long time, which may increase the irreversible capacity of the battery.

(2) Discharge precautions:

When installing and installing equipment, pay attention to the overcurrent protection function of lithium batteries. When the current used by the device exceeds the maximum output current of the battery, the overcurrent protection circuit will activate, causing the device to malfunction

(3) Usage environment:

The charging temperature of the battery should be between 0 ℃ and 45 ℃, and the discharging temperature should be between -20 ℃ and 60 ℃. Batteries should be stored in a clean, dry, and ventilated environment, avoiding contact with corrosive substances, and kept away from sources of fire and heat.

(4) Battery storage:

Long term idle lithium iron phosphate batteries are recommended to be charged to a level of 40% to reduce capacity loss. Newly purchased lithium iron phosphate batteries also need to be charged before use to ensure battery performance and lifespan.

Following the above suggestions can effectively extend the lifespan of lithium iron phosphate batteries and ensure safe use.

(5) Battery maintenance:

1. Perform a full charge and discharge cycle once a week: It is recommended to conduct at least one full charge and discharge cycle per week, where the battery goes from fully depleted to fully charged, which helps calibrate the battery's battery level indicator and ensure its health status.

2. Control charging time and temperature: If the battery is not fully charged after charging for more than 10 hours, stop charging immediately and check the temperature. If the battery overheats, it should be sent for repair in a timely manner. In cases where immediate repair is not possible, the total charging time should be controlled to not exceed 8 hours to prevent battery expansion and deformation.

3. Avoid overcharging and discharging: Lithium iron phosphate batteries have no memory effect and can theoretically be charged at any time. However, to avoid rapid changes in internal pressure, it is recommended to determine the charging frequency based on usage. At the same time, avoid excessive discharge of the battery to avoid affecting its performance.

4. Choose the appropriate charger: Prioritize using the original charger provided by the factory and avoid using incompatible chargers to ensure safe and efficient charging.

5. Pay attention to the usage environment: Lithium iron phosphate batteries are suitable for charging in an environment between 0 ℃ and 45 ℃, and discharging in an environment between -20 ℃ and 60 ℃. The environment should be kept clean, dry, ventilated, and away from corrosive substances, fire sources, and heat sources.

6. Long term storage recommendation: If the battery needs to be idle for a long time, it is recommended to charge it to 40% battery level to reduce capacity loss.

(6) Safe transportation

When transporting lithium iron phosphate batteries, a series of safety regulations and standards need to be followed to ensure the safety and compliance of the transportation process. Here are some key precautions:

1. Packaging requirements: Lithium iron phosphate batteries must be packaged in accordance with the regulations for the transportation of dangerous goods.

Temperature and humidity control: When charging and storing the battery pack, attention should be paid to temperature and humidity control. When charging, the temperature should be between 0-50 ℃ and the humidity should be within the range of 45-85% RH. Exceeding this range may result in decreased battery performance or safety issues. 3. Safety measures: The battery pack should be stored separately and avoid contact with metal or sharp objects to prevent short circuits or damage. During transportation, keep away from heat sources, high pressure, avoid hitting, and prevent children from playing with it.

## 9. Lubricating oil maintenance of gearbox

| Item | worm gear oil(220) |
|---|---|
| kinematic viscosit 40℃ mm2/s | 198-242 |
| Viscosity index ≥ | 90 |
| Flash point (opening)℃ ≥ | 180 |

| | |
|---|---|
| copper corrosion(1000℃,3h) | 1 |
| pour point ℃ ≤ | -18 |
| Corrosion test | Rustless |

The maintenance of gearbox oil mainly includes the replacement, inspection, precautions for use, and handling in special circumstances of lubricating oil.

**Oil replacement:**

1. Gearboxes generally use turbine worm lubricating oil with a viscosity of 220CS. After running for 7-14 days (150-300 hours), engine oil should be added, and the maximum limit of the oil pot should not be exceeded when adding engine oil.

2. During use, it is necessary to regularly check the oil quality and replace deteriorated oil products that contain impurities, contamination, or have decomposed or aged at any time. In general, gearboxes that work continuously for a long time can have their oil replaced every 6-8 months, while gearboxes that work for no more than 8 hours a day can have their oil replaced every 10-12 months.

3. The new oil replaced must have the same brand number as the original oil used, and different grades and types of oil should not be mixed.

**Precautions for using engine oil:**

1. Use fixed grade gear oil, different grades of oil cannot be mixed.

2. When refueling, pay attention to the amount of oil to ensure that the gearbox is lubricated correctly.

3. For reducers with heavy loads, frequent starts, and poor operating environments, some lubricating oil additives can be selected to ensure that the gear oil still adheres to the gear surface when the reducer stops running, forming a protective film.

**Handling in special circumstances:**

If high oil temperature or abnormal noise is found, the machine should be stopped for inspection in a timely manner, and the fault should be eliminated before continuing to use.

2. When the ambient temperature is below 0 ° C, the lubricating oil must be heated to above 0 ° C or low freezing point lubricating oil must

be used before starting.

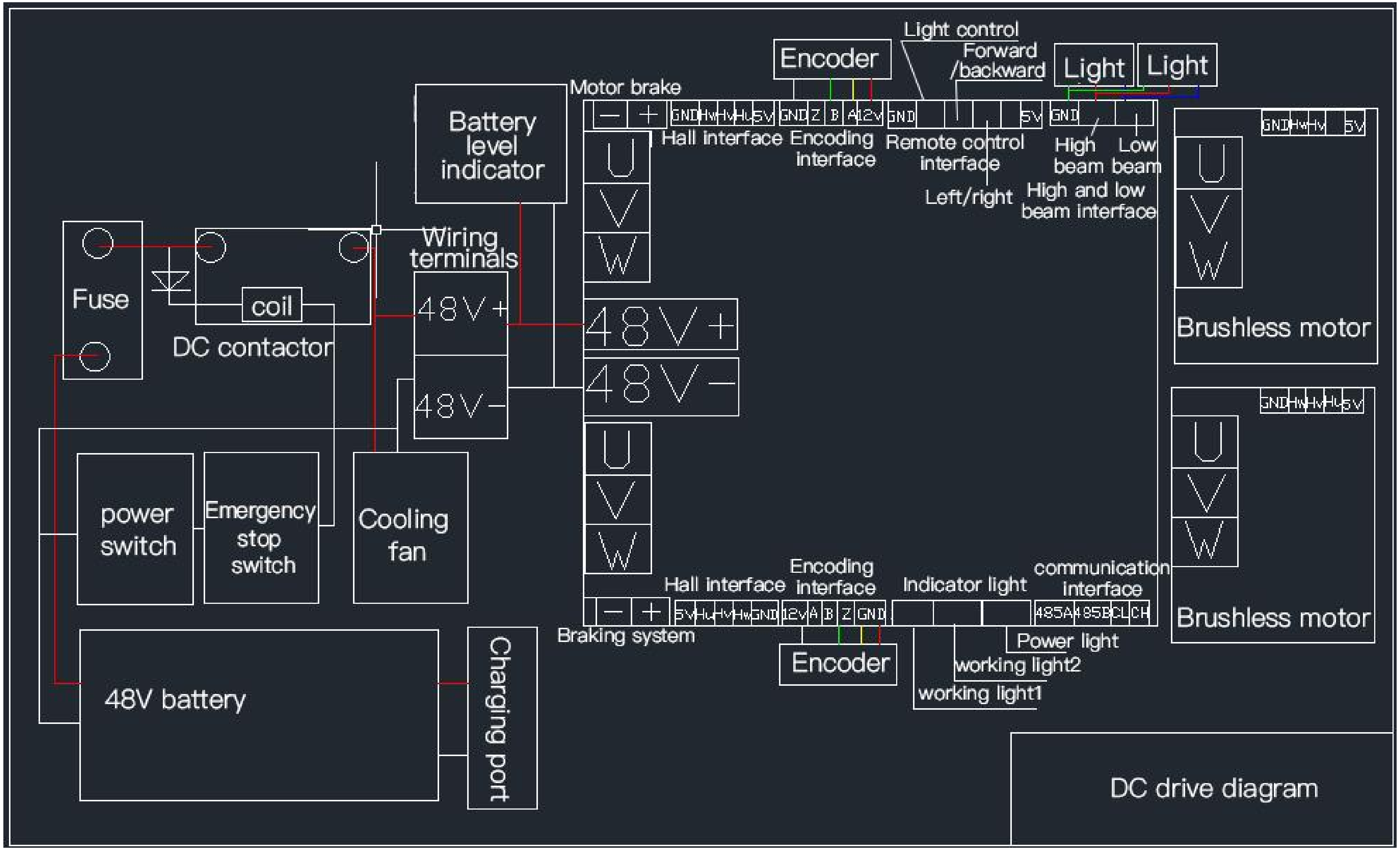
Through the above maintenance content, the normal operation of the worm gear reducer can be ensured, the service life can be extended, and the maintenance cost can be reduced.

When refueling, pay attention to the amount of oil to ensure that the gearbox is lubricated correctly.

For reducers with heavy loads, frequent starts, and poor operating environments, some lubricating oil additives can be selected to ensure that the gear oil still adheres to the gear surface when the reducer stops running, forming a protective film.

**10.Wiring diagram**

See below diagram



Battery level indicator
Motor brake
Encoder
Light control
Forward /backward
Light
Light
GND Hw Hv Hu 5V
GND Z B A 12v
GND
5V
GND
Hall interface
Encoding interface
Remote control interface
Left/right
High beam
Low beam
High and low beam interface
GND Hw Hv 5V
Fuse
DC contactor
coil
Wiring terminals
48V+
48V-
48V+
48V-
U
V
W
Brushless motor
GND Hw Hv Hu 5V
Brushless motor
power switch
Emergency stop switch
Cooling fan
48V battery
Charging port
Hall interface
Encoding interface
Indicator light
communication interface
5V Hu Hv Hw GND
12v A B Z GND
485A 485B CL CH
Braking system
Encoder
Power light
working light2
working light1
DC drive diagram

## 11. Communication Protocol

Reference standard 485 Modbus RTU protocol CRC16 check

Serial port parameters

Baud rate: 38400

Data bits: 8

Parity: no parity

Stop bit: 1

Slave fixed ID: 01

Control mode

1. T10 wireless remote control mode
2. Modbus speed closed loop control mode
3. Modbus position closed loop control mode
4. 485 joystick control mode

Register address and description

| **Register function** | **address code (Decimal)** | **address code (Hexadecimal)** | **Read and write function code** | **Register value** |
|---|---|---|---|---|
| 485 rocker (vertical) | 10 | 0x0A | 0x03,0x06,0x10 | Signed hexadecimal parameter<br>Range: 800~~-800 |

| | | | | |
|---|---|---|---|---|
| 485 joystick (horizontal) | 11 | 0x0B | 0x03,0x06,0x10 | Signed hexadecimal parameter<br>Range: 800~~-800 |
| Left wheel target speed | 12 | 0x0C | 0x03,0x06,0x10 | Signed hexadecimal parameter |
| Right wheel target speed | 13 | 0x0D | 0x03,0x06,0x10 | Signed hexadecimal parameter |
| Control mode | 80 | 0x50 | 0x03,0x06,0x10 | 0: wireless remote control mode<br>1: Speed closed loop mode<br>2. Position closed loop mode |
| Start and stop instructions | 81 | 0x51 | 0x03,0x06,0x10 | 0: stop, parameter configuration state<br>1: Start |
| Left wheel target speed | 82 | 0x52 | 0x03,0x06,0x10 | Signed hexadecimal parameter |
| Right wheel target speed | 83 | 0x53 | 0x03,0x06,0x10 | Signed hexadecimal parameter |
| Left wheel target mileage H | 84 | 0x54 | 0x03,0x06,0x10 | 32-bit unsigned pulse number |
| Left wheel target mileage L | 85 | 0x55 | 0x03,0x06,0x10 | |
| Right wheel target mileage H | 86 | 0x56 | 0x03,0x06,0x10 | 32-bit unsigned pulse number |
| Right wheel target mileage L | 87 | 0x57 | 0x03,0x06,0x10 | |
| Actual speed of Left wheel | 54 | 0x36 | 0x03 | Signed hexadecimal parameter |
| Actual speed of right wheel | 55 | 0x37 | 0x03 | Signed hexadecimal parameter |
| Actual mileage of Left wheel H | 56 | 0x38 | 0x03 | 32-bit unsigned pulse number |
| Actual mileage of Left wheel L | 57 | 0x39 | 0x03 | |
| Actual mileage of right wheel H | 58 | 0x40 | 0x03 | 32-bit unsigned pulse number |
| Actual mileage of right wheel L | 59 | 0x41 | 0x03 | |

| | | | | |
|---|---|---|---|---|
| Displacement loop control status | 60 | 0x3C | 0x03 | State when detecting the displacement ring<br>1. 0x01: running<br>2. 0x00: Before or after running |
| High and low beam control | 61 | 0x3D | 0x03,0x06 | 1. 0x01 Turn off the lights<br>2. 0x02 low beam<br>3. 0x03 high beam |
| Radar 1 | 62 | 0x3E | 0x03 | Range: 25-250cm<br>0: Sensor failure<br>255: Accessible or not detected |
| Radar 2 | 63 | 0x3F | 0x03 | Range: 25-250cm<br>0: Sensor failure<br>255: Accessible or not detected |
| Radar 3 | 64 | 0x40 | 0x03 | Range: 25-250cm<br>0: Sensor failure<br>255: Accessible or not detected |
| Radar 4 | 65 | 0x41 | 0x03 | Range: 25-250cm<br>0: Sensor failure<br>255: Accessible or not detected |
| Radar 5 | 66 | 0x42 | 0x03 | Range: 25-250cm<br>0: Sensor failure<br>255: Accessible or not detected |

| | | | | |
|---|---|---|---|---|
| Radar 6 | 67 | 0x43 | 0x03 | Range: 25-250cm<br>0: Sensor failure<br>255: Accessible or not detected |
| Radar 7 | 68 | 0x44 | 0x03 | Range: 25-250cm<br>0: Sensor failure<br>255: Accessible or not detected |
| Radar 8 | 69 | 0x45 | 0x03 | Range: 25-250cm<br>0: Sensor failure<br>255: Accessible or not detected |
| Radar Control Mode | 70 | 0x46 | 0x03,0x06,0x10 | 1.0: Cancel the radar control mode<br>1: Start the radar control mode |
| Error code output | 71 | 0x47 | 0x03 | 1: 0x00: normal without abnormalities |
| Device ID | 512 | 0x200 | 0x03,0x06 | 1:0x01--0xFE( 1-255)<br>0xff is the broadcast address |

1. 485 interface remote control mode
   1. Two-dimensional joystick data input (X, Y) corresponds to register address (0x0A, 0X0B)
   2. Input range: (800----(-800)) origin (0,0)
   3. The refresh rate is not more than 1 second interval (recommended 50MS)
   4. If the data interaction interval exceeds 1S, the program executes the communication interruption judgment mechanism

5. Feedback real-time speed linear speed (5600---(-5600)) positive and negative sign represents direction

6. Speed feedback value (left wheel, right wheel) corresponding register (0x36, 0x37)

2. Speed closed-loop control mode

1. Speed data input (L left wheel, R right wheel) corresponds to the register address (0X0C, 0X0D)

2. Input range: (5600---(-5600)) origin (0,0)

3. The refresh rate is not more than 1 second interval (recommended 50MS)

4. If the data interaction interval exceeds 1S, the program executes the communication interruption judgment mechanism

5. Feedback real-time speed line speed 0-5600 line speed, the speed has no positive or negative distinction

6. Speed feedback value (left wheel, right wheel) corresponding register (0x36, 0x37)

Device ID range: 0x01--0xFE(1-254) 0XFF is the broadcast address

When the user forgets the ID, the slave ID of the device can be retrieved through the broadcast address

Speed closed-loop control:

Unit: line/second

Accuracy: 20 lines

Range: 600-5600 (forward is positive input, backward is negative input) minimum speed is 600 lines/second

Configuration parameters: speed loop mode, speed value real-time input update

Write input speed frequency: 1HZ (continuous input) Stop when data is interrupted, and the speed can be adjusted during driving

Position closed-loop control:

Speed parameters are the same as speed closed loop

Displacement parameters:

unit: line

Range: 200-100000 lines

Accuracy: 100 lines

Configuration parameters: displacement loop mode, speed value, displacement value, operation start-stop mode operation after update

After the end of the displacement loop, the real-time displacement value is cleared, and the parameters need to be repeatedly configured before executing the vehicle control. During the driving process, the displacement and speed cannot be modified, and the start-stop function can be controlled in real time.

Read input displacement frequency 1hz

Note: After the parameter is updated, after writing the start command, the start-stop command key needs to be read at a reading frequency of 1hz. It is used to support the heartbeat command of the displacement loop movement. If it exceeds, the stop will be cleared.

The speed of the product is unified according to line/second. According to the configuration of our company, the actual mileage of each encoder revolution is 0.51m (fixed value) Line/second is converted into m/S formula (collection speed/number of encoder lines)*0.51

The actual speed is related to the number of lines of the encoder

Note: Radar sensor is the default function (customer optional)